



Tribhuvan University
Faculty of Humanities and Social Sciences

A
PROJECT PROPOSAL
ON
NEPAL BUS TICKET BOOKING SYSTEM
(Course Code: CAPJ356)



Submitted to Department of Information Technology
Kabhre Multiple Campus

*A Proposal submitted for the partial fulfillment of requirements for the degree of
Bachelor in Computer Application (Sixth Semester)*

Submitted by

Nischal Shrestha

Registration Number: 6-2-244-15-2023

Roll No: 24402165

Under Supervision of

[Supervisor Name]

October, 2026

Tribhuvan University
Faculty of Humanities and Social Sciences
Kabhre Multiple Campus

Supervisor's Recommendation

I hereby recommend that this project proposal prepared under my supervision by **Nischal Shrestha** entitled “**NEPAL BUS TICKET BOOKING SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the proposal defense.

.....
Signature
[Supervisor Name]
SUPERVISOR
[Academic Designation]
Department of Information Technology
Kabhre Multiple Campus, Banepa-Budol, Nepal

Letter of Approval

This project proposal entitled “**NEPAL BUS TICKET BOOKING SYSTEM**” by **Nischal Shrestha** as a partial fulfillment of the requirements for the degree of Bachelor in Computer Application (Sixth Semester) has been examined by us and is accepted.

Panel of Examiners:

Name

Signature

Date

1. HOD [HOD Name]

2.

3.

4.

Abstract

Long-distance bus ticketing in Nepal still depends largely on physical counters, phone calls, and local agents. This causes problems for passengers, who cannot easily check seat availability or compare fares, and for operators, who manage seat records manually and face double booking and disputes. This project proposes the Nepal Bus Ticket Booking System, a web-based application that lets passengers search for buses by origin, destination, and travel date, choose seats from a visual seat map, and book tickets online. Bus operators can manage routes, buses, schedules, and fares, and administrators can oversee users, bookings, and reports.

The system prevents double booking through a temporary seat-locking mechanism. It uses sorting and searching algorithms to present results by price or departure time, a fare calculation module that supports special pricing rules, and reports on bookings, revenue, and seat occupancy. Online payment is integrated through eSewa, and a booking is confirmed only after the payment is verified. The system will be developed as a responsive web application using Next.js, NestJS, PostgreSQL, and Tailwind CSS.

Keywords: bus ticket booking, seat locking, web application, Nepal, NestJS, PostgreSQL

Acknowledgement

I would like to express my sincere gratitude to my supervisor, [Supervisor Name], for the guidance, suggestions, and encouragement given throughout the preparation of this project proposal.

I am thankful to [HOD Name], Head of the Department of Information Technology, and to Kabhre Multiple Campus for giving me the opportunity to work on this project. I also thank all the faculty members of the department for the knowledge and support they have provided during my studies.

Finally, I would like to thank my family and friends for their continuous support and motivation.

Nischal Shrestha
Registration Number: 6-2-244-15-2023
Roll No: 24402165
October, 2026

Table of Contents

ABSTRACT	i
ACKNOWLEDGEMENT	ii
LIST OF ABBREVIATIONS	iv
LIST OF TABLES	v
CHAPTER 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Scope and Limitation	2
1.4.1 Scope	2
1.4.2 Limitations	3
1.5 Development Methodology	3
1.6 Report Organization	4
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW	5
2.1 Background Study	5
2.1.1 Study of Existing System	5
2.1.2 Key Concepts	5
2.2 Literature Review	6
CHAPTER 3: SYSTEM ANALYSIS AND DESIGN	8
3.1 System Analysis	8
3.1.1 Requirement Analysis	8
3.1.2 Feasibility Analysis	10
3.1.3 Process Modelling	11
3.1.4 Data Modelling	11
3.2 High-Level Design	11
CHAPTER 4: IMPLEMENTATION AND TESTING	15
4.1 Implementation	15
4.1.1 Tools Used	15
4.1.2 Implementation Details of Modules	15
4.2 Testing Plan	15
4.3 Project Schedule	16
CHAPTER 5: EXPECTED OUTCOME AND FUTURE RECOMMENDATIONS	17
5.1 Expected Outcome	17
5.2 Future Recommendations	17
REFERENCES	18

List of Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
CASE	Computer-Aided Software Engineering
CSS	Cascading Style Sheets
GPS	Global Positioning System
HOD	Head of Department
HTML	HyperText Markup Language
IEEE	Institute of Electrical and Electronics Engineers
PHP	PHP: Hypertext Preprocessor
SMS	Short Message Service
SQL	Structured Query Language
XHTML	Extensible HyperText Markup Language

List of Tables

2.1	Comparison of the manual system, existing platforms, and the proposed system	6
-----	--	---

CHAPTER 1: INTRODUCTION

1.1 Introduction

Road transport is the main means of long-distance travel in Nepal, and buses connect cities such as Kathmandu, Pokhara, Chitwan, and Birgunj to the rest of the country. Despite this, ticket booking for many bus services still depends on physical counters, phone calls, or local agents. Passengers often have to visit the counter in person to check seat availability, compare prices, or confirm a booking, and operators must manage seat records manually.

The Nepal Bus Ticket Booking System is a web-based application that allows passengers to search for buses, view available seats, and reserve tickets online. Passengers can search by origin, destination, and travel date, select seats from a visual seat map, and receive a booking confirmation. Bus operators can manage their routes, buses, schedules, and fares, while administrators oversee users, bookings, and reports.

The system focuses on solving the problem of double booking through a seat-locking mechanism, which holds a seat temporarily while a passenger completes the booking. It uses sorting and searching algorithms to present results by price or departure time, and a fare calculation module for pricing rules. The system will be developed as a responsive, mobile-friendly web application using Next.js, NestJS, PostgreSQL, and Tailwind CSS. Online payment will be integrated through eSewa.

1.2 Problem Statement

Many long-distance bus services in Nepal still rely on manual ticketing. Passengers must visit counters or contact agents to find out which buses are running, whether seats are available, and what the fare is. Comparing the options of different operators on a route takes time, and a passenger often cannot be sure a seat is confirmed until they reach the counter.

Manual booking also creates problems for operators. Seat records are kept on paper or in separate registers, which makes errors likely. When two agents or counters sell the same seat, double booking occurs and leads to disputes at boarding time. Cancellations, fare changes during festival seasons, and sales reports are hard to track without a central system.

Because of these issues, there is a need for a system that shows live seat availability, prevents the same seat from being booked twice, and gives operators one place to manage routes, schedules, fares, and bookings. This project addresses that need by developing

a web-based bus ticket booking system tailored to the Nepali context.

1.3 Objectives

The main objective of this project is to develop a web-based bus ticket booking system that makes seat reservation easier and more reliable for passengers and operators in Nepal.

The specific objectives are:

1. To develop a search module that lets passengers find buses by origin, destination, and travel date, and sort the results by price or departure time.
2. To implement a seat selection and booking module that shows live seat availability and prevents double booking through a temporary seat-locking mechanism.
3. To provide role-based access for passengers, bus operators, and administrators, with secure authentication.
4. To develop a fare calculation module that supports route-based pricing and special pricing rules, such as peak-season fares.
5. To implement a booking and cancellation workflow with online payment through eSewa, confirming a booking only after the payment is verified.
6. To generate reports for operators and administrators, such as total bookings, revenue, and seat occupancy.

1.4 Scope and Limitation

1.4.1 Scope

The scope of this project covers the design and development of a web-based bus ticket booking system for long-distance routes in Nepal. The system is a responsive web application for desktop and mobile browsers and includes the following:

1. Three user roles (passengers, bus operators, and administrators) with registration, login, secure password storage, and role-based access.
2. Management of routes, buses, schedules, and fares by operators and administrators.
3. Searching for buses by origin, destination, and travel date, with results sorted by price or departure time.
4. A visual seat map showing available and booked seats, with temporary seat locking to prevent double booking.

5. Fare calculation with special rules such as peak-season fares, along with booking, cancellation, and booking history with online payment through eSewa.
6. Reports for operators and administrators, such as total bookings, revenue, and seat occupancy.

1.4.2 Limitations

The system has the following limitations:

1. Payment works only in the eSewa test (sandbox) environment with test accounts, so no real money is processed. Moving to live payments needs a registered merchant account and is left for future work.
2. The system runs on sample data. It is not connected to real bus operators, and the routes and fares are for demonstration only.
3. It does not provide live bus tracking or GPS location.
4. It is a web application only. A native mobile app is not developed.
5. It is not tested for large-scale traffic and is designed for a limited number of users.
6. Notifications such as SMS or email are not included in the current version.

1.5 Development Methodology

The system is developed using the iterative incremental model of the software development life cycle. Instead of building everything at once, the system is divided into increments. Each increment goes through requirements, design, implementation, and testing, and ends with a working part of the system that is reviewed with the supervisor before the next increment starts. This model suits a single developer with weekly supervisor meetings, because problems in the design are found early, while they are still cheap to fix.

The planned increments are:

1. Authentication and role-based access for passengers, operators, and administrators.
2. Management of buses, routes, schedules, and fares.
3. Bus search and the seat map, with sorting of results.
4. Seat locking and booking, including cancellation.
5. Online payment through eSewa, with payment verification.
6. Special fare rules and reports.

Each increment is tested when it is finished, and a final round of system testing and documentation follows the last increment.

1.6 Report Organization

This proposal is organized into five chapters. Chapter 1 introduces the project, the problem, the objectives, the scope and limitations, and the development methodology. Chapter 2 presents the background study, including a study of the existing system, and the literature review. Chapter 3 covers the system analysis and design: the requirements, the feasibility analysis, the object-oriented models, the high-level design, and the algorithms. Chapter 4 describes the planned implementation and testing, which covers the tools, the modules, the testing approach, and the project schedule. Chapter 5 gives the expected outcome and the future recommendations. The references are listed at the end.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

2.1.1 Study of Existing System

Manual system. In the traditional system, a passenger books a long-distance bus ticket by visiting the operator's counter, calling the office, or going through a local agent. The staff check a paper register or a seat chart, tell the passenger which seats are free, take the payment in cash, and write the booking by hand. Each counter or agent keeps its own record, so seat information is not shared in real time. This leads to double booking, long queues, errors in the records, and difficulty in producing sales reports.

Existing online platforms. Several online platforms already offer bus ticket booking in Nepal. BusSewa describes itself as a real-time booking platform that lets passengers search routes, compare fares, choose seats, and receive e-tickets, with payment through wallets, banks, and cards [1]. GoMyGo allows several passengers to be booked in one transaction, sends confirmations by email and SMS, and follows each operator's own cancellation policy [2]. Okay Journey offers an interactive seat map and a choice of bus class [3]. Trip Turbo lets users filter buses by price and departure time, pick window or aisle seats from a seat map, and receive an e-ticket by email and SMS [4].

Comparison. Table 2.1 compares the manual system, the existing online platforms, and the proposed system.

Summary. Existing platforms have solved the main problems of the manual system for the operators they work with. This project does not aim to replace them. Its purpose is to build the core booking logic as a complete system with its own modules: seat locking to prevent double booking, fare rules, sorting and searching of results, role-based management for passengers, operators, and administrators, and reports. The modules are written for this project and not taken from plugins or third-party services, so they can be studied, tested, and extended.

2.1.2 Key Concepts

[To be written]

Table 2.1: Comparison of the manual system, existing platforms, and the proposed system

Feature	Manual system	Existing online platforms	Proposed system
Seat availability	Checked at the counter or by phone	Real-time	Real-time
Seat selection	Done by staff, limited	Interactive seat map	Interactive seat map
Booking record	Paper registers	Stored in a database	Stored in a database
Payment	Cash at the counter	Wallets, banks, cards	eSewa (sandbox test mode)
Notifications	None	Email and SMS	Not included
Operator reports	Hard to prepare	Offered through the platform	Bookings, revenue, occupancy
Scope	Single counter	Commercial, many operators	Academic project, sample data

2.2 Literature Review

Several studies have looked at online bus ticket reservation, and their basic features are very similar.

Oloyede et al. developed a web-based bus reservation system for a transport service in Nigeria using XHTML, PHP, SQL, Ajax, CSS, and JavaScript [5]. Their system lets users check seat availability, reserve a ticket, and cancel it, and it gives administrators the means to verify users, add vehicles, and generate reports [5]. They report that manual reservation handled separately at each branch causes delays and risks of duplicate bookings [5]. Their work, however, does not describe how two simultaneous requests for the same seat are handled.

Chavan et al. describe a computerized reservation system to replace manual record keeping, which they say caused errors in data entry and disputes with customers [6]. The system keeps passenger details, seat availability, price per seat, and bill generation, and is built with HTML, PHP, CSS, JavaScript, and MySQL [6]. It is a short study of a basic design and does not cover fare rules or concurrent bookings.

Venkat and Raju present a bus ticketing application built with Django and MySQL [7]. Users search by source, destination, and date, and seat availability is calculated in real time from the booking records [7]. The system also offers fare calculation, booking history, cancellation of future tickets, and an administrative module for routes and bookings [7]. The abstract states that data consistency is maintained for multiple users but does not describe the mechanism used [7].

The seat-conflict problem is treated in more detail in ticketing research for high traffic. Zhang et al. studied a railway ticketing system and note that under heavy load the same ticket can be sold more than once [8]. Their design temporarily locks the seat when an order is created and deducts it only after payment succeeds, and uses an atomic token-deduction step in Redis to prevent overselling [8]. Their system uses microservices and several middleware components, which is far larger than what a single-developer project needs [8].

Summary of the review. The reviewed systems agree on a common core: searching buses, checking seats, booking, cancelling, and administrator management [5, 6, 7]. The simpler systems do not explain how double booking is prevented, while the research on high-traffic ticketing shows that it is the central problem [8]. This project keeps the common core and adds an explicit temporary seat-locking mechanism, role-based management for passengers, operators, and administrators, and fare rules. It also takes a lighter approach than the microservice design in [8], which suits a single-developer project on a single server.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

The purpose of system analysis and design is to systematically create effective and efficient solutions to meet specific business or organizational needs[cite: 2]. System analysis is conducted to study a system and its components in order to identify core objectives, specify functional requirements, and define system boundaries[cite: 2].

3.1 System Analysis

The proposed Nepal Bus Ticket Booking System will be built following the structured approach to software development. The structured approach provides a step-by-step, process-driven methodology that focuses on data flow, system decomposition, and structured data modeling. System Analysis will evaluate the operational, functional, and technical requirements of the system using tools such as Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and Data Dictionaries to ensure a clear separation of data and process logic before implementation.

3.1.1 Requirement Analysis

Requirement analysis defines the functional and non-functional requirements of the proposed system. Figure 3.1 shows the Context Diagram (Level 0 DFD) of the Nepal Bus Ticket Booking System, which establishes the system boundary and models high-level interactions between external entities and the system.

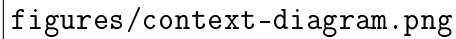
The system identifies four primary external entities:

- **Passenger:** Searches for bus trips, selects seats with temporary holds, initiates online payments, receives booking confirmations, and requests cancellations.
- **Bus Operator:** Configures fleet details, seat layouts, routes, schedules, dynamic fare rules, and views booking reports for their fleet.
- **Administrator:** Approves new bus operators, manages user accounts, and monitors system-wide analytics.
- **eSewa Gateway:** External payment gateway integrated to verify transactions securely.

i. Functional Requirements

Passenger

1. Register, log in, and log out securely.

The image is a placeholder for a Context Diagram (Level 0 DFD) of the Nepal Bus Ticket Booking System. It contains the text "figures/context-diagram.png" centered within a rectangular frame.

figures/context-diagram.png

Figure 3.1: Context Diagram (Level 0 DFD) of Nepal Bus Ticket Booking System

2. Search for buses by origin, destination, and travel date, and sort results by price or departure time.
3. View the seat map of a bus and select available seats.
4. Enter passenger details and pay online through eSewa.
5. View booking confirmations and booking history.
6. Cancel a booking according to system cancellation rules.

Bus Operator

1. Log in to an operator account.
2. Add and manage buses, including bus types and seat layouts.
3. Add and manage routes, schedules, and fares (including peak-season pricing).
4. View bookings and reports for their own buses.
5. Cancel scheduled trips when necessary.

Administrator

1. Log in to an administrator account.
2. Approve and manage bus operators.

3. Manage user accounts, including account suspension.
4. View all system-wide bookings and reports.

System

1. Prevent double booking by holding selected seats temporarily during checkout.
2. Calculate fares based on base route pricing and active fare rules.
3. Confirm bookings only after successful payment verification with eSewa.
4. Enforce role-based access control across all features.

ii. Non-Functional Requirements

1. *Security*: Store passwords in hashed form; restrict access using role-based permissions.
2. *Data Integrity*: Ensure atomic transactions for bookings, seat locks, and payments.
3. *Usability*: Responsive and simple interface accessible via mobile and desktop browsers.
4. *Performance*: Rapid seat map rendering and search result response times.
5. *Maintainability*: Modular codebase structured for future extensibility.
6. *Compatibility*: Full support across modern web browsers.

3.1.2 Feasibility Analysis

A feasibility study determines whether the proposed system can be successfully developed and deployed using available technical skills, time, and resources.

3.1.2.1 Technical Feasibility

The system will be built using open-source technologies: Next.js and Tailwind CSS for the user interface, NestJS for the server (written in TypeScript), and PostgreSQL for relational data storage. PostgreSQL's support for ACID transactions and row-level locking directly enables the seat-locking mechanism required to prevent concurrent double bookings. Payment integration will utilize the eSewa ePay sandbox environment for secure transaction processing. Git and GitHub will handle version control. The technical risk involved in seat concurrency will be mitigated using database row-level locking during simultaneous booking requests. Therefore, the project is technically feasible.

3.1.2.2 Operational Feasibility

The proposed system replaces physical ticket counters and phone calls with a streamlined online workflow: search, select seat, and pay. Passengers require only a modern web browser and internet connection. Operators and administrators receive dedicated web-based dashboards to manage fleets, routes, schedules, and analytics, reducing paper-based record-keeping. The system is operationally feasible.

3.1.2.3 Economic Feasibility

All core frameworks and development tools (Next.js, NestJS, PostgreSQL, Tailwind CSS, Git) are free and open-source, eliminating software licensing costs. Payment processing during development utilizes the free eSewa sandbox testing suite. The primary resource investment is development time, while operational benefits include reduced labor overhead, automated seat management, and improved record accuracy. Thus, the system is economically feasible.

3.1.3 Process Modelling

Process modelling illustrates how data moves through the system and how internal processes transform inputs into outputs.

Level 1 Data Flow Diagram (DFD). Figure 3.2 shows the Level 1 DFD, detailing the core functional processes of the system including User Authentication, Trip Search, Seat Hold & Allocation, Payment Processing, and Management Dashboards.

3.1.4 Data Modelling

Data modelling defines the logical data structures and entity relationships within the database.

Entity-Relationship Diagram (ERD). Figure 3.4 illustrates the Entity-Relationship Diagram for the system. It models core entities—including User, Bus, Route, Trip, Seat, Booking, and Payment—along with their attributes, primary keys, foreign key constraints, and cardinalities.

3.2 High-Level Design

Figure ?? presents the high-level system architecture structured in three main layers alongside an external payment integration.

The web frontend (Next.js and Tailwind CSS) acts as the client layer for passengers, operators, and administrators. The backend server (NestJS) processes REST API re-

figures/level1-dfd.pdf

Figure 3.2: Level 1 Data Flow Diagram of Nepal Bus Ticket Booking System

figures/erd-diagram.pdf

Figure 3.3: Entity-Relationship Diagram (ERD) of the system

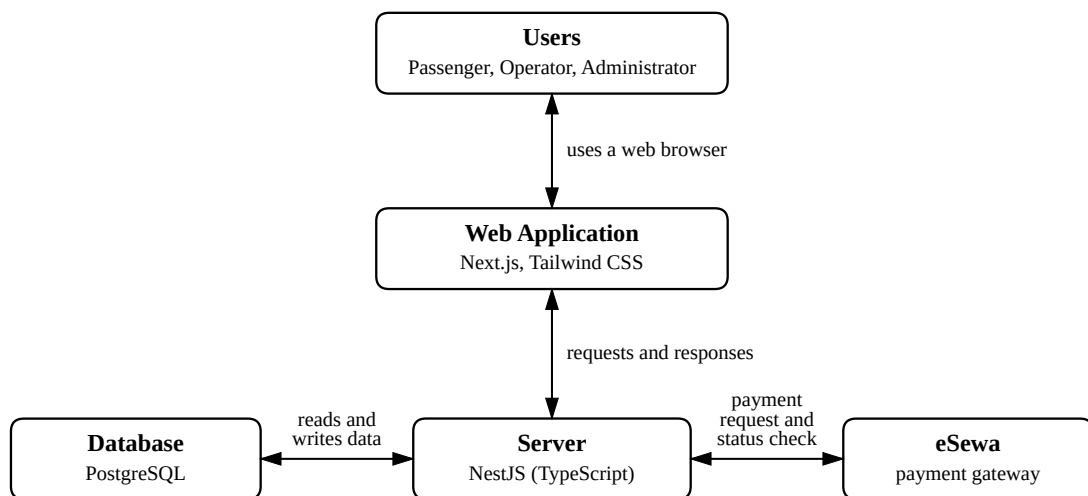


Figure 3.4: High-Level Architecture of the System

quests, handles authentication, applies fare rules, manages seat locking logic, and communicates with PostgreSQL for database operations. Online payment verification is managed asynchronously via eSewa API calls.

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

The following tools are used to develop and manage the project.

CASE and development tools. Visual Studio Code is the code editor. The use case diagram, ER diagram, Data Flow Diagrams (DFDs), system flowchart, and architecture diagram are created using modeling tools. Claude is used as an AI assistant for design discussion, drafting, and review. Notion is used for project tracking and management. Git is used for version control, and the source code is hosted on GitHub.

Programming languages and frameworks. The system is written in TypeScript on Node.js. Next.js and Tailwind CSS are used for the front end, and NestJS is used for the back end.

Database. PostgreSQL is used as the database.

4.1.2 Implementation Details of Modules

The system is divided into the following modules, which follow the increments described in Section 1.5.

1. *Authentication and role management:* registration, login, and role-based access for passengers, operators, and administrators.
2. *Bus, route, and trip management:* buses with seat layouts, routes, schedules, and fares.
3. *Search and seat map:* searching buses by origin, destination, and date, with results sorted by price or departure time, and a seat map for each trip.
4. *Booking and seat locking:* temporary seat holds, booking, and cancellation.
5. *Payment:* online payment through eSewa, with verification before a booking is confirmed.
6. *Fare rules and reports:* special fare rules and reports on bookings, revenue, and seat occupancy.

4.2 Testing Plan

The system will be tested as each increment is completed. The detailed test cases will be written during development and reported in the final report.

Unit testing. Unit tests will cover the functions of each module, such as fare calculation, seat holding and release, sorting of search results, and password hashing.

System testing. System tests will follow the complete booking flow, from search to confirmed booking, including payment verification and cancellation. They will include a test with simultaneous requests for the same seat, to confirm that double booking does not occur.

4.3 Project Schedule

[To be written]

CHAPTER 5: EXPECTED OUTCOME AND FUTURE RECOMMENDATIONS

5.1 Expected Outcome

At the end of the project, the following are expected to be delivered:

1. A working web-based bus ticket booking system with separate access for passengers, bus operators, and administrators, protected by secure login.
2. A search module that finds buses by origin, destination, and travel date, and sorts the results by price or departure time.
3. A seat selection and booking module with a live seat map and a temporary seat-locking mechanism that prevents double booking, tested with simultaneous booking requests.
4. A fare calculation module that supports route-based pricing and special pricing rules, such as peak-season fares.
5. A booking, cancellation, and booking-history workflow with online payment through eSewa, where a booking is confirmed only after the payment is verified.
6. Reports for operators and administrators on bookings, revenue, and seat occupancy.

In addition, the project will deliver complete documentation, including the requirement analysis, design diagrams, and test results, and the source code managed in a Git repository. The system will be a working application, demonstrated with sample data.

5.2 Future Recommendations

The system can be extended in the following ways:

1. Live payments through eSewa with a registered merchant account, and other payment options.
2. Email and SMS notifications for booking confirmation and changes to a trip.
3. Live bus tracking using GPS.
4. A native mobile application.
5. Testing and tuning for a large number of simultaneous users.
6. Analytics that help operators decide on schedules and fares, such as demand and occupancy trends.

REFERENCES

- [1] BusSewa, “Nepal’s Trusted Online Ticketing & Travel Platform.” [Online]. Available: <https://bussewa.com.np/>. [Accessed: Oct. 4, 2026].
- [2] GoMyGo, “Book Bus Tickets Online in Nepal.” [Online]. Available: <https://gomygo.com/>. [Accessed: Oct. 4, 2026].
- [3] Okay Journey, “Nepal’s Leading Trusted Online Bus Ticket Platform.” [Online]. Available: <https://www.okayjourney.com/>. [Accessed: Oct. 4, 2026].
- [4] Trip Turbo, “How to Book Bus Tickets Online Using Trip Turbo in Nepal,” Mar. 9, 2025. [Online]. Available: <https://tripturbo.com/blog/how-to-book-bus-tickets-online/>. [Accessed: Oct. 4, 2026].
- [5] M. O. Oloyede, S. M. Alaya, and K. S. Adewole, “Development of an online bus ticket reservation system for a transportation service in Nigeria,” *Computer Engineering and Intelligent Systems*, vol. 5, no. 12, pp. 9–17, 2014.
- [6] M. Chavan *et al.*, “Online bus reservation system,” *Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol.*, vol. 7, no. 3, pp. 494–496, May–Jun. 2021.
- [7] Thotakura Jai Sai Venkat and A. Naga Raju, “A web-based online bus ticketing application for efficient public transport reservation and management,” *Int. J. Eng. Res. Sci. Technol.*, vol. 22, no. 2, pp. 1677–1688, Apr. 2026.
- [8] Z. Zhang, X. Zhang, and X. Li, “Securing high-concurrency ticket sales: A framework based on microservice,” arXiv:2512.24941, Dec. 2025. [Online]. Available: <https://arxiv.org/pdf/2512.24941>